IN5340/IN9340 Project III

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Code: `github.com/Twaal/Statistical_Signal_Processing_IN5340/tree/master/Project_3`

Agenda

Problem statement

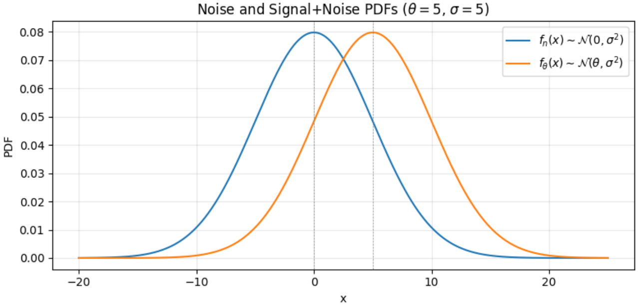
- Binary detection with additive Gaussian noise:

$$H_0 : x = n, \qquad H_1 : x = \theta + n, \quad n \sim \mathcal{N}(0, \sigma^2)$$

- Use $\theta = 5$ and $\sigma = 5$ as baseline case.
- Design and evaluate:
  - CFAR detector for given P_{FA}
  - ROC curve (theoretical vs empirical)
  - Improvement using sample mean of N measurements

b) Noise and signal+noise PDFs

- Under H_0: $x \sim \mathcal{N}(0, \sigma^2)$
- Under H_1: $x \sim \mathcal{N}(\theta, \sigma^2)$
- For $\theta/\sigma = 1$, distributions overlap strongly.
- Detection is possible, but not highly reliable in this regime.

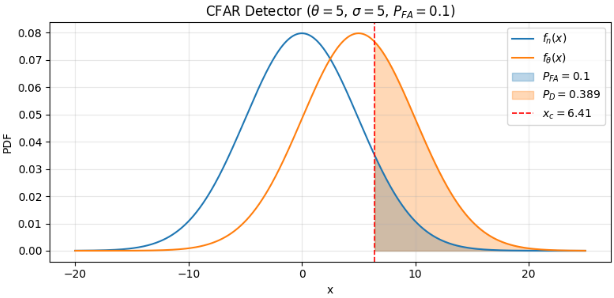


c) CFAR detector with $P_{FA} = 0.1$

$$x_c = \sigma Q^{-1}(P_{FA}) = 5\, Q^{-1}(0.1) \approx 6.41$$

$$P_D = Q\left(\frac{x_c - \theta}{\sigma}\right) \approx 0.389$$

- Blue shaded area: false-alarm probability P_{FA}.
- Orange shaded area: detection probability P_D.
- Dashed red line: threshold x_c.

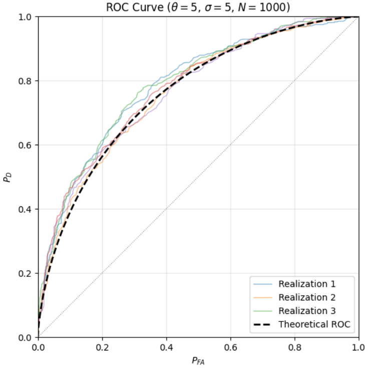


d) ROC empirical and theoretical

- Data model:

  $$s \in \{0, \theta\},\ P(s = 0) = P(s = \theta) = 0.5,\ x$$

- Empirical ROC computed by threshold sweep on simulated data.
- Multiple realizations vary, but cluster around the theoretical ROC.



e) Multi-sample detector, minimum N

Using sample mean of N measurements:

$$\bar{x}|H_0 \sim \mathcal{N}\left(0, \frac{\sigma^2}{N}\right), \qquad \bar{x}|H_1 \sim \mathcal{N}\left(\theta, \frac{\sigma^2}{N}\right)$$

For CFAR with requirements $P_{FA} = 0.01$ and $P_D > 0.9$:

$$N \geq \left(\frac{\sigma}{\theta}\right)^2 \left(Q^{-1}(P_{FA}) + Q^{-1}(1 - P_D)\right)^2$$

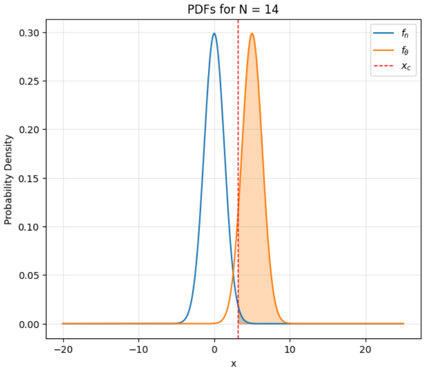
Numerically:

$$N_{\text{min, cont}} \approx 13.02 \quad \Rightarrow \quad N_{\text{min}} = 14$$

At $N = 14$: $x_c \approx 3.11$ and $P_D \approx 0.922$.

f) Improved detector, $N = 14$

- Variance shrinks from σ^2 to σ^2/N.
- The two PDFs become better separated.
- This gives better detection performance for the same P_{FA}.

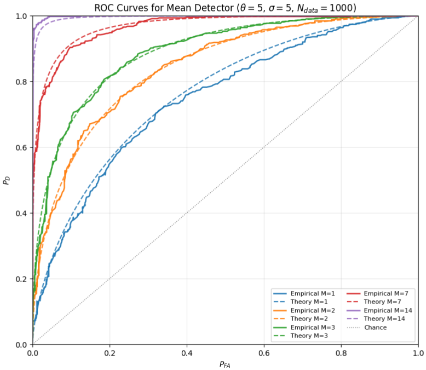


g) ROC for multiple measurements

From stdout, single realization:

- $M = 1$: $\mathrm{AUC}_{emp} = 0.745$, $\mathrm{AUC}_{th} = 0.760$
- $M = 2$: $\mathrm{AUC}_{emp} = 0.843$, $\mathrm{AUC}_{th} = 0.841$
- $M = 3$: $\mathrm{AUC}_{emp} = 0.888$, $\mathrm{AUC}_{th} = 0.890$
- $M = 7$: $\mathrm{AUC}_{emp} = 0.965$, $\mathrm{AUC}_{th} = 0.969$
- $M = 14$: $\mathrm{AUC}_{emp} = 0.998$, $\mathrm{AUC}_{th} = 0.996$

ROC shifts toward top-left as M increases.

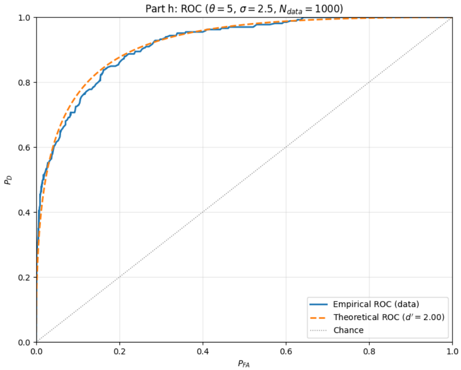


h) Imbalanced priors, $\theta = 5$, $\sigma = 2.5$

From stdout:

- Generated prevalence: $\pi_1 = 0.266$, $\pi_0 = 0.734$
- $d' = \theta/\sigma = 2.000$
- $\mathrm{AUC}_{emp} = 0.918$, $\mathrm{AUC}_{th} = 0.921$

The empirical ROC still matches theory well.

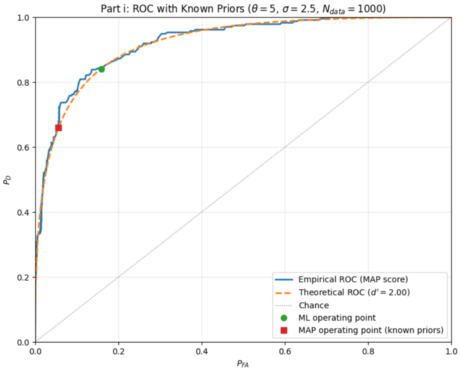


i) Known priors, MAP operating point

From stdout:

- ML threshold: $y_t = 2.500$
- MAP threshold: $y_t = 3.968$
- ML point: $(P_{FA}, P_D) = (0.159, 0.841)$
- MAP point: $(P_{FA}, P_D) = (0.056, 0.660)$

MAP reduces false alarms when H_1 is rare.

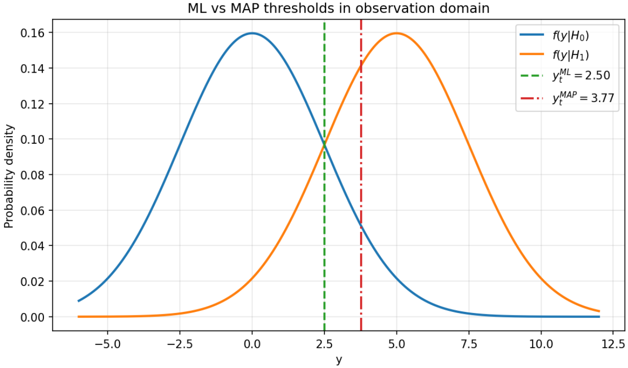


i) PDFs with ML and MAP thresholds

Interpretation:

- ML ignores class priors and uses the midpoint threshold.
- MAP accounts for rare H_1 and shifts threshold right.
- This shift reduces false alarms, but it also lowers detection probability.

$$x_t^{ML} = \frac{\theta}{2}, \qquad x_t^{MAP} = \frac{\theta}{2} + \frac{\sigma^2}{\theta} \log \frac{P_0}{P_1}$$



For the Gaussian shift model:

$$y|H_0 \sim \mathcal{N}(0, \sigma^2), \qquad y|H_1 \sim \mathcal{N}(\theta, \sigma^2)$$

$$f_0(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}}, \qquad f_1(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-\theta)^2}{2\sigma^2}}$$

$$\log \frac{f_1(y)}{f_0(y)} = -\frac{(y-\theta)^2}{2\sigma^2} + \frac{y^2}{2\sigma^2} = \frac{\theta}{\sigma^2} y - \frac{\theta^2}{2\sigma^2}$$

MAP rule with equal costs:

$$\Pr(H_1 \mid y) \underset{H_0}{\overset{H_1}{\gtrless}} \Pr(H_0 \mid y)$$

$$\frac{f_1(y)}{f_0(y)} \underset{H_0}{\overset{H_1}{\gtrless}} \frac{\Pr(H_0)}{\Pr(H_1)} = \frac{P_0}{P_1}$$

$$\Rightarrow \quad y \underset{H_0}{\overset{H_1}{\gtrless}} y_t^{MAP} = \frac{\theta}{2} + \frac{\sigma^2}{\theta} \log \frac{P_0}{P_1}$$

i) Priors, MAP, and ML

- Priors enter only through the term $\log(P_0/P_1)$ in the MAP threshold.
- If H_1 is rare ($P_1 < P_0$), then $\log(P_0/P_1) > 0$ and threshold increases.
- Higher threshold means fewer false alarms, but also fewer detections.

ML is a special case of MAP:

$$P_0 = P_1 \Rightarrow \log \frac{P_0}{P_1} = 0 \Rightarrow y_t^{MAP} = \frac{\theta}{2} = y_t^{ML}$$

ML assumes equal priors. MAP extends the same rule when prior probabilities are known.

Code snippets, essential parts

1) Multi-measurement ROC model (g):

```
d_prime_n = theta * np.sqrt(M) / sigma
pd_theory = norm.sf(norm.isf(pfa_grid) - d_prime_n)
```

2) MAP score with known priors (i):

```
llr = (theta/sigma**2)*y - theta**2/(2*sigma**2)
score_map = llr + np.log(P1/P0)
```

3) MAP threshold in observation domain:

$$y_t^{MAP} = \frac{\theta}{2} + \frac{\sigma^2}{\theta} \log \frac{P_0}{P_1}$$

Summary

- For a single measurement with $\theta = \sigma = 5$, overlap is substantial and P_D is modest at practical P_{FA}.
- CFAR with $P_{FA} = 0.1$ gives threshold $x_c \approx 6.41$ and $P_D \approx 0.389$.
- ROC simulations agree with theory in all experiments, with AUC differences of only a few hundredths.
- Averaging measurements strongly improves performance, from AUC 0.745 at $M = 1$ to 0.998 at $M = 14$.
- With imbalanced priors, MAP shifts the operating point to reduce false alarms from 0.159 to 0.066, with lower P_D.